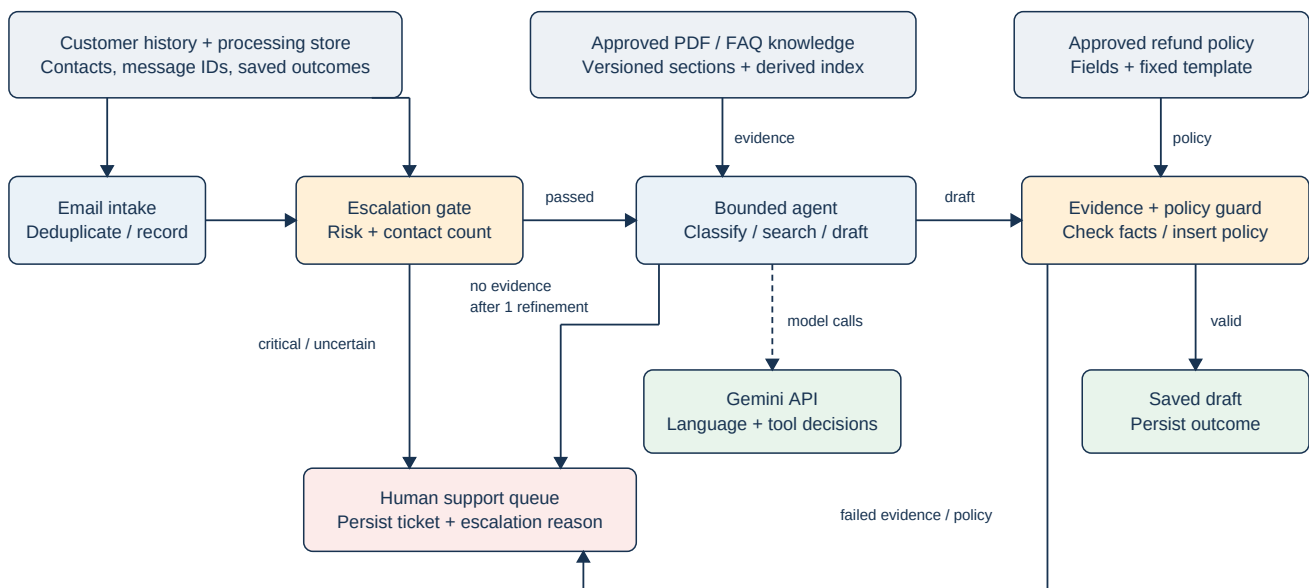


Agentic Architect Challenge

Jerence | Developer Intern assessment | Python + Gemini

Part 1 | Customer-support email architecture



All stages log IDs, decisions, source versions, latency and outcomes. Solid: workflow/data. Dashed: model dependency.

Part 1: proposed design and safety boundaries

Flag data loss, service outage, security breach, or more than three contacts in seven days before drafting. Unavailable history or unresolved risk requires human handling. Assumptions: the current email counts, the UTC window is inclusive, and retries share a message ID. The store records contacts and terminal outcomes atomically. Critical emails can be classified separately without drafting. The bounded agent may refine a knowledge query once. Refund terms come from approved versioned fields/templates inserted by code; missing or conflicting policy blocks the response.

Part 2: scraping and summarisation

The hypothetical naive script misses JavaScript and sends noisy, excessive text in one prompt. The implemented replacement extracts main/article content and tables, renders sparse pages with Playwright, and caps HTML at 2 MiB. Up to 100 source chunks are summarised; repeated reduction keeps content payloads within 6,000 characters. A shortening pass plus a hard cap enforce 120 words. Long jobs cost more calls/time. Over-budget pages fail explicitly; authentication and infinite scrolling are outside scope.

Part 3: document agent

The implemented agent supplies the complete fictional OrbitDesk handbook to Gemini and retains twelve completed turns. Gemini selects calculator calls in AUTO mode; Python returns restricted arithmetic results. Two tool calls and three model requests bound each question. Document answers require a real section and matching excerpt; missing evidence is withheld. Reset clears memory.

Trade-offs, failures and verification

Direct SDK calls keep the implementation small; one short document needs no vector database. Conservative human handoffs reduce automation coverage, and fixed refund templates require policy maintenance. Gemini has a 30-second per-attempt timeout and up to two total attempts. Failures produce explicit outcomes; failed chat turns are not saved. Logs omit prompts and keys. Offline tests cover orchestration, guardrails and real JavaScript rendering; separate live cases check model behaviour. See docs/verification.md for results. Citation matching does not prove semantic correctness. Part 1 is a proposed design; no production load benchmark is claimed. Measure latency, quotas and queue age before choosing concurrency.